

DRY-RUN GUIDE | 01_majority_element.py
Source: Python\dsa_solutions\01_majority_element.py

CORE BEHAVIOR

Defined callable(s): main(). No input() read is visible. Output statements include:
print(candidate if numbers.count(candidate) > count // 2 else -1).

REPRESENTATIVE INPUT

Function call: main(); use small arguments matching the source parameter types.

STEP TRACE

Call main() and follow its assignments, branches, and loop updates. A return statement gives the result to the caller; this module has no detected main entry point.

EXPECTED OUTPUT / RETURN

Evaluate the listed print expression(s) for the assumed values: print(candidate if numbers.count(candidate) > count // 2 else -1). Prompts may also appear.

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\01_majority_element.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     values = list(map(int, sys.stdin.read().split()))
006 |     if not values:
007 |         return
008 |     count = values[0]
009 |     numbers = values[1:1 + count]
010 |     candidate = None
011 |     votes = 0
012 |     for number in numbers:
013 |         if votes == 0:
014 |             candidate = number
015 |             votes += 1 if number == candidate else -1
016 |     print(candidate if numbers.count(candidate) > count // 2 else -1)
017 |
018 |
019 | if __name__ == "__main__":
020 |     main()
021 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.